Mutual Fund Analytics Platform

Bluestock Fintech Capstone — Final Project Report

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KushmaReddy/Mutual_Fund_Analytics: Bluestock Capstone Project

Executive Summary

The Mutual Fund Analytics Platform is an end-to-end data analytics project built for the Bluestock Fintech Capstone Program. It covers 40 mutual fund schemes across 10 fund houses, processing ten raw datasets — fund master, NAV history (46,000+ records), investor transactions (32,000+ records), scheme performance, portfolio holdings, benchmark indices, industry AUM, and SIP inflows — through a Python ETL pipeline into a structured SQLite database (bluestock_mf.db).

Performance analytics computed CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown for every scheme. Advanced analytics extended this with Historical VaR/CVaR (95%), a 90-Day Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and Sector Concentration (HHI). A rule-based recommendation engine ranks schemes by Sharpe Ratio within a chosen risk grade, and a four-page Power BI dashboard presents the findings interactively.

Headline results: ICICI Pru Midcap Fund posted the highest CAGR (32.83%) and strong Alpha (0.295); Mirae Asset Large Cap Fund delivered the best risk-adjusted return (Sharpe 1.45, Sortino 2.39); SBI Small Cap Fund – Direct Plan carried the deepest drawdown (-52.57%) and weakest VaR profile. Most critically, SIP Continuity Analysis flagged 1,332 of 1,362 tracked investors (97.8%) as “At Risk” due to an average 65-day gap between contributions — a major retention signal for the business.

This report documents the data sources, ETL pipeline, EDA findings, performance analysis, dashboard, business recommendations, and known limitations of the platform.

1. Data

1.1 Data Sources

Ten raw datasets were used, covering scheme master data, historical NAV, investor transactions, scheme performance, portfolio holdings, benchmark indices, industry AUM, and SIP inflows.

Dataset	Description	Approx. Records
Fund Master	40 schemes, 10 fund houses — category, sub-category, expense ratio, fund manager, risk grade	40
NAV History	Daily NAV per scheme, 2022–2026	46,000+
Investor Transactions	SIP / Lumpsum / Redemption with investor demographics	32,000+
Scheme Performance	Pre-computed 1/3/5-yr returns, Sharpe, expense ratio	40
Portfolio Holdings	Sector allocation per scheme (used for HHI)	320
Benchmark Indices	NIFTY 50 / Midcap 150 / SmallCap TRI (used for Alpha/Beta)	—
AUM / SIP Inflows	Fund-house AUM and category-wise monthly inflows, 2022–2025	90 / 144

1.2 Data Dictionary (key fields)

File	Column	Description
fund_master.csv	amfi_code, fund_house, scheme_name, category, sub_category, risk_grade	Scheme identity, classification and risk level
nav_history.csv	amfi_code, nav_date, nav, daily_return	Daily NAV and computed daily return

File	Column	Description
investor_transactions.csv	transaction_id, investor_id, transaction_type, amount, transaction_date, state, kyc_status	Investor-level transaction log
scheme_performance.csv	amfi_code, expense_ratio, sharpe_ratio, returns_1y/3y/5y	Pre-computed scheme performance metrics

Full column-level definitions are maintained in the repository's data_dictionary.md, along with the cleaning rules applied to each field (duplicate removal, datetime conversion, NAV/amount validation, KYC status checks, missing-value handling).

1.3 Data Quality Snapshot

Before analysis, every dataset was profiled for shape, missing values, duplicates, and type consistency (data_quality_report.py, day1_data_quality_report.md). AMFI code validation confirmed 40 codes in Fund Master and 40 in NAV History, with zero missing codes — full referential integrity between the two tables.

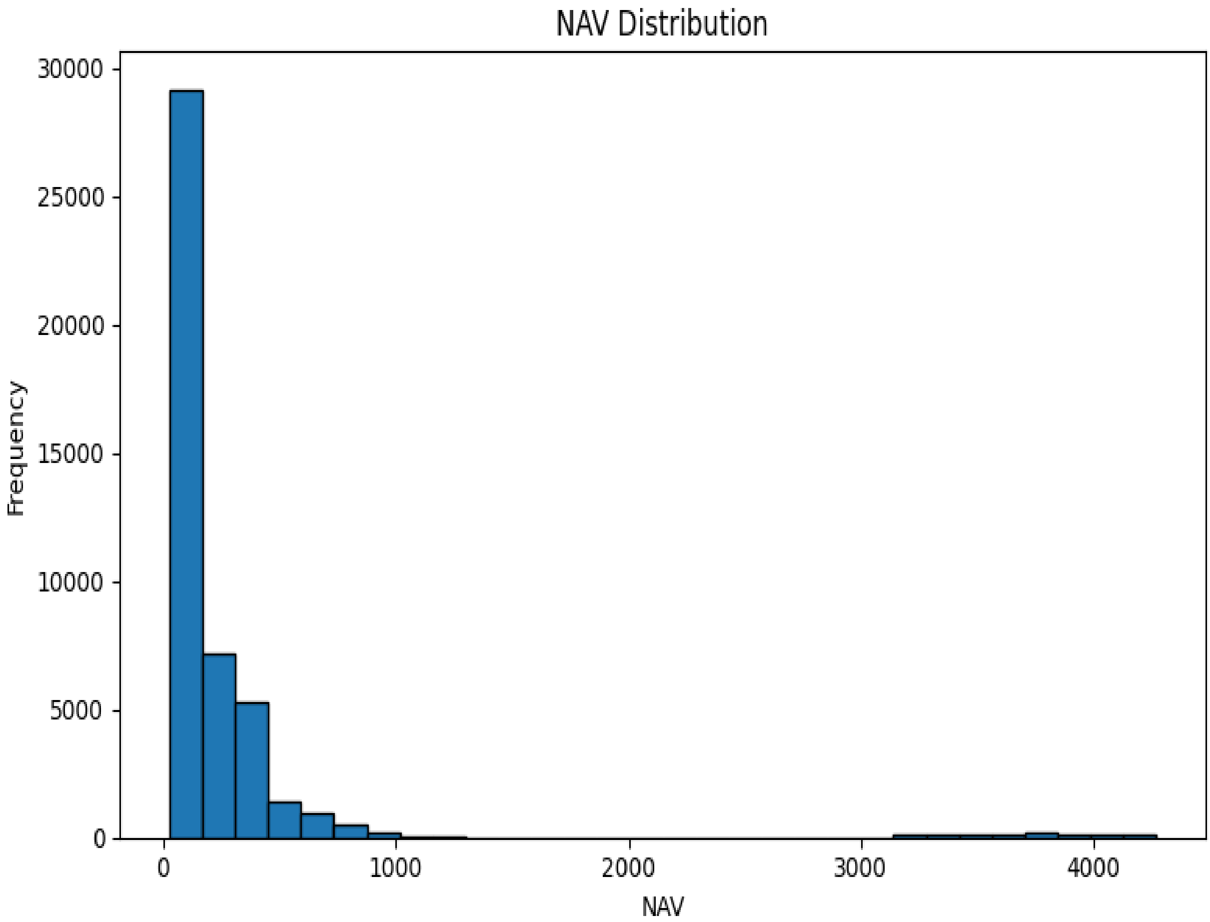


Figure 1.1 – NAV Distribution across all schemes and dates

2. ETL (Extract, Transform, Load)

2.1 Pipeline Overview

Raw CSVs (data/raw) were extracted with Pandas, transformed through cleaning scripts, and loaded into a SQLite database (bluestock_mf.db) with four tables — dim_fund, fact_nav, fact_transactions, and fact_performance — joined on amfi_code.

Key transformations

- Duplicate rows removed via df.drop_duplicates() across all ten datasets.
- Date columns converted to datetime; return and expense-ratio columns coerced to numeric.
- Missing NAV values forward-filled per scheme: df.groupby('amfi_code')['nav'].ffill().
- Transaction type standardized (str.strip().str.title()); KYC status and expense-ratio ranges validated.
- AMFI codes cross-validated between Fund Master and NAV History — 40 codes matched on both sides, zero missing.

example: NAV cleaning (clean_nav_history.py)

```
df["date"] = pd.to_datetime(df["date"])
df = df.sort_values(by=["amfi_code", "date"])
df["nav"] = df.groupby("amfi_code")["nav"].ffill()
df = df.drop_duplicates()
df.to_csv("data/processed/clean_nav.csv", index=False)
```

2.2 Database Schema

Table	Purpose
dim_fund	Scheme master — fund house, category, expense ratio, risk grade
fact_nav	Daily NAV history per scheme
fact_transactions	Investor-level SIP / Lumpsum / Redemption records
fact_performance	Return, Alpha, Beta, Sharpe, Sortino, Max Drawdown, AUM

Row counts across all four tables were verified after every load (check_database.py), confirming the pipeline completed end-to-end with full referential integrity.

2.3 Eample SQL Analysis

SQL was used for filtering, aggregation, and joins across the four tables. Representative queries (see queries.sql for the full set of 10):

```
-- Top 5 Funds by AUM
SELECT scheme_name, aum_crore FROM fact_performance
JOIN dim_fund ON fact_performance.amfi_code = dim_fund.amfi_code
ORDER BY aum_crore DESC LIMIT 5;

-- Number of Investors by City Tier
SELECT city_tier, COUNT(investor_id) AS total_investors
FROM fact_transactions GROUP BY city_tier ORDER BY total_investors DESC;
```

3. EDA Findings

Exploratory analysis covered NAV distribution, NAV trends, AUM growth, category inflows, and investor transaction patterns using Pandas, NumPy, Matplotlib and Seaborn.

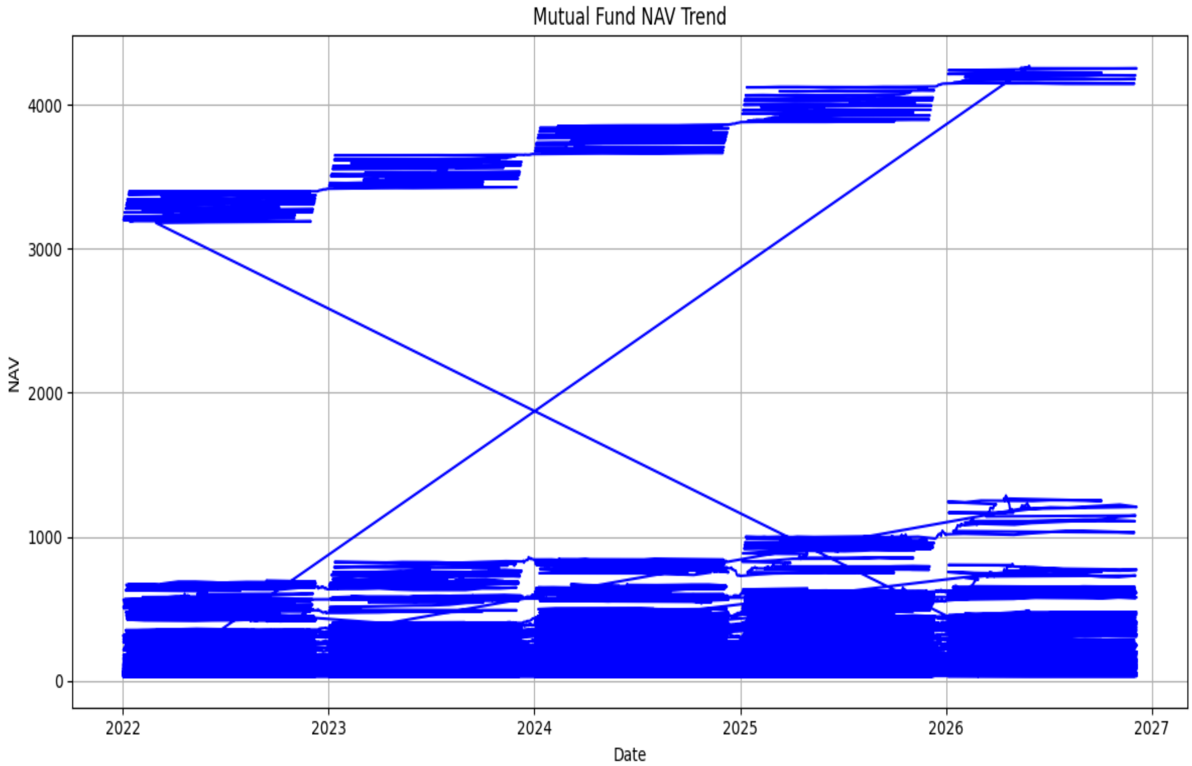


Figure 3.1 – Historical NAV Trend, all 40 schemes (2022–2026)

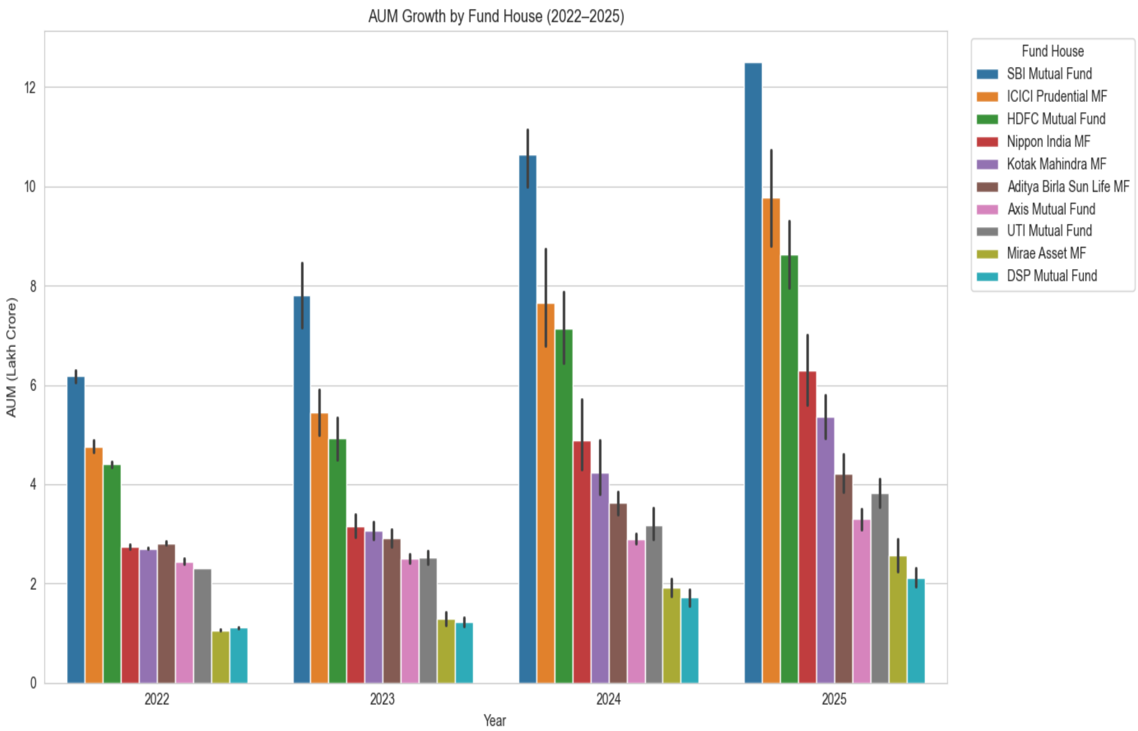


Figure 3.2 – AUM Growth by Fund House (2022–2025)

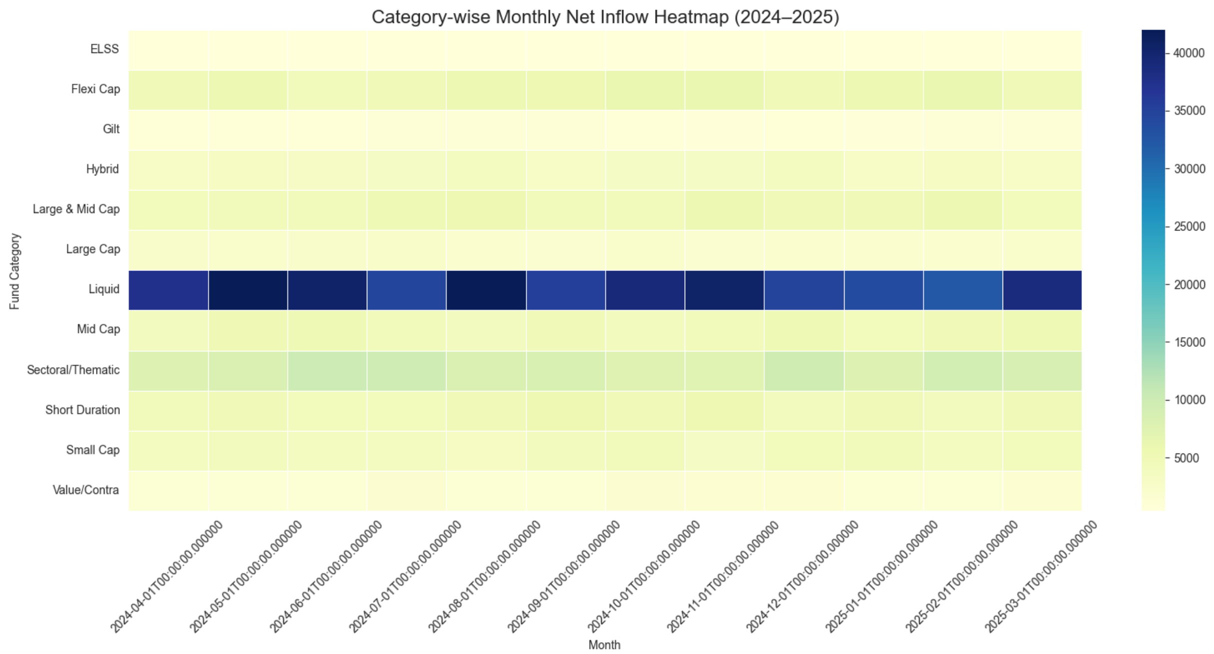


Figure 3.3 – Category-wise Monthly Net Inflow Heatmap

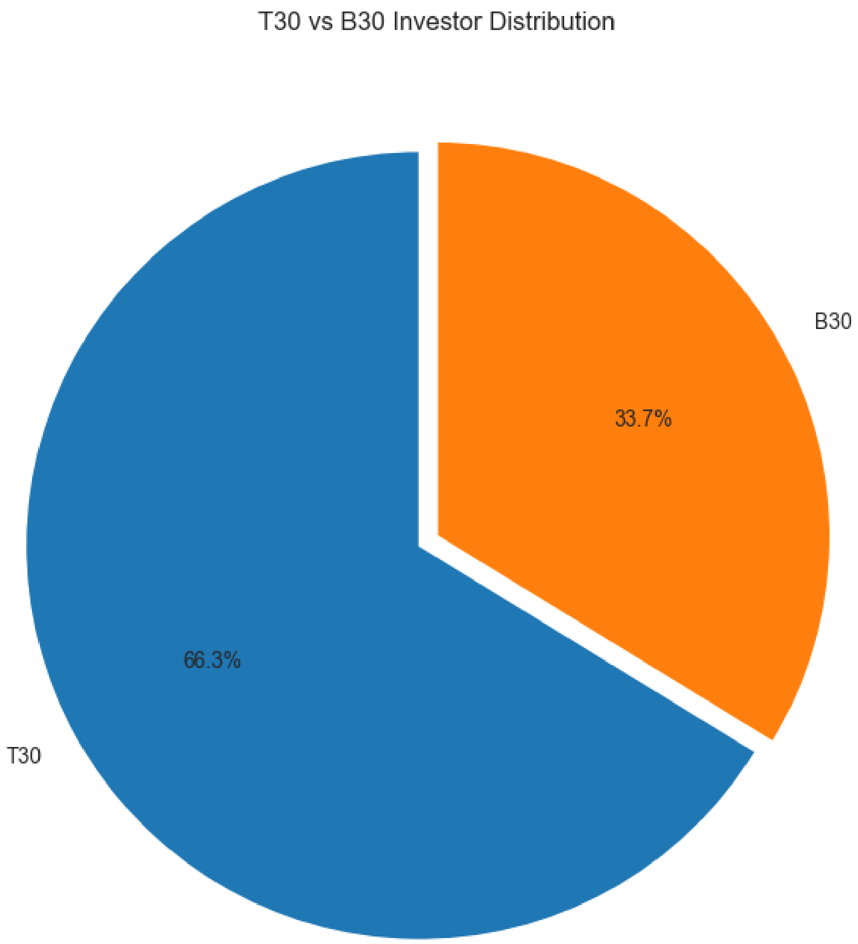


Figure 3.4 – Investor Distribution by City Tier (T30 vs B30)

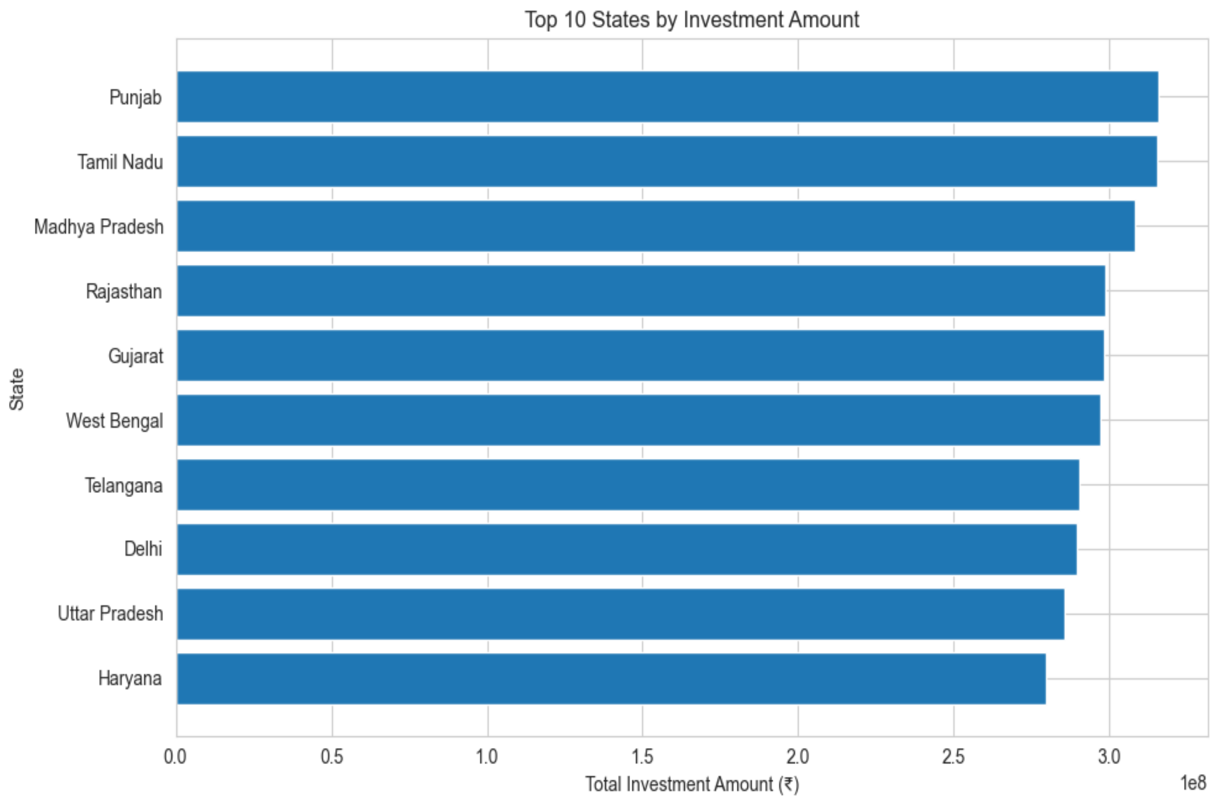


Figure 3.5 – Top 10 States by Investment Amount

Key Findings

- NAV shows a small set of high-NAV outlier schemes (>₹3,000) skewing the overall distribution; most schemes trade well under ₹1,000.
- SBI Mutual Fund leads industry AUM every year 2022–2025 (₹6.3L cr → ₹12.5L cr), ahead of ICICI Prudential MF and HDFC Mutual Fund.
- Liquid Funds dominate monthly category inflows by a wide margin — a short-term cash-parking pattern rather than long-term growth allocation.
- SIP transactions account for 58.5% of investor activity, ahead of Redemptions (35.3%) and Lumpsum (6.2%).
- Average SIP ticket size is stable across age groups (₹10.9K–₹11.6K), with a mild skew toward the 56+ bracket.
- Punjab, Tamil Nadu, and Madhya Pradesh are the top three states by transaction value.

4. Performance Analysis

4.1 Return & Risk-Adjusted Metrics

CAGR, Annual Return, Sharpe Ratio, Sortino Ratio, Alpha, Beta and Maximum Drawdown were computed for all 40 schemes (04_performance_analytics.ipynb).

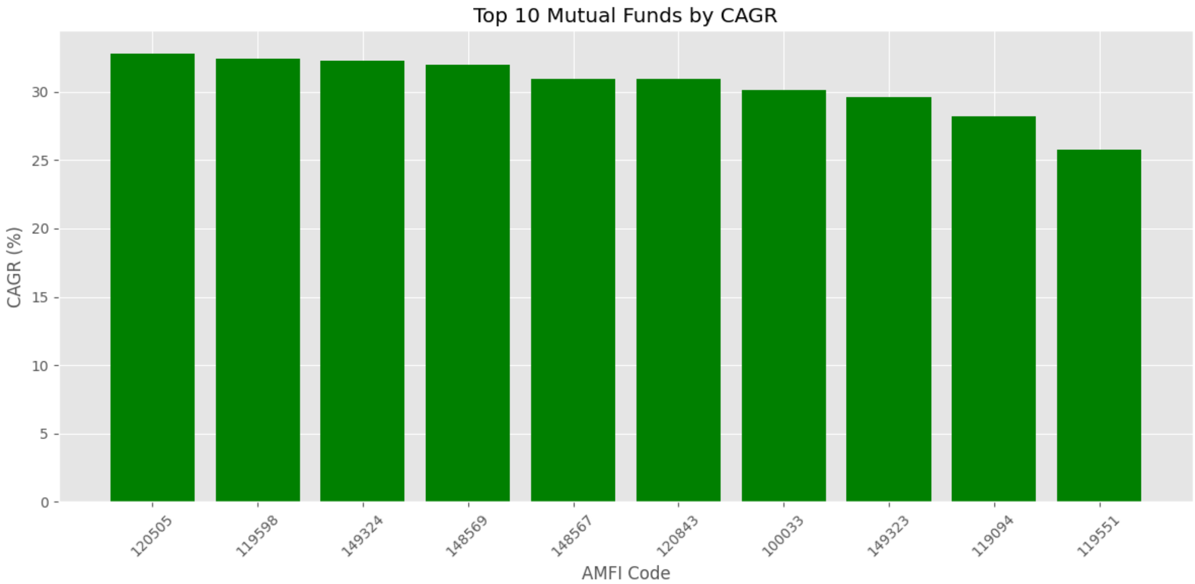


Figure 4.1 – Top 10 Mutual Funds by CAGR

Metric	Leader	Value
Highest CAGR	ICICI Pru Midcap Fund	32.83%
Highest Alpha (among CAGR leaders)	ICICI Pru Midcap Fund	0.295
Highest Sharpe Ratio	Mirae Asset Large Cap Fund	1.45
Highest Sortino Ratio	Mirae Asset Large Cap Fund	2.39
Deepest Max Drawdown (highest risk)	SBI Small Cap Fund – Direct Plan	-52.57%

- Small/Mid Cap equity schemes dominate the top of the CAGR table (28–33%); Debt/Liquid schemes are far more stable but single-digit.
- A high Sharpe Ratio does not always coincide with the highest CAGR — Mirae Asset Large Cap Fund leads risk-adjusted return despite ranking outside the CAGR top 5.

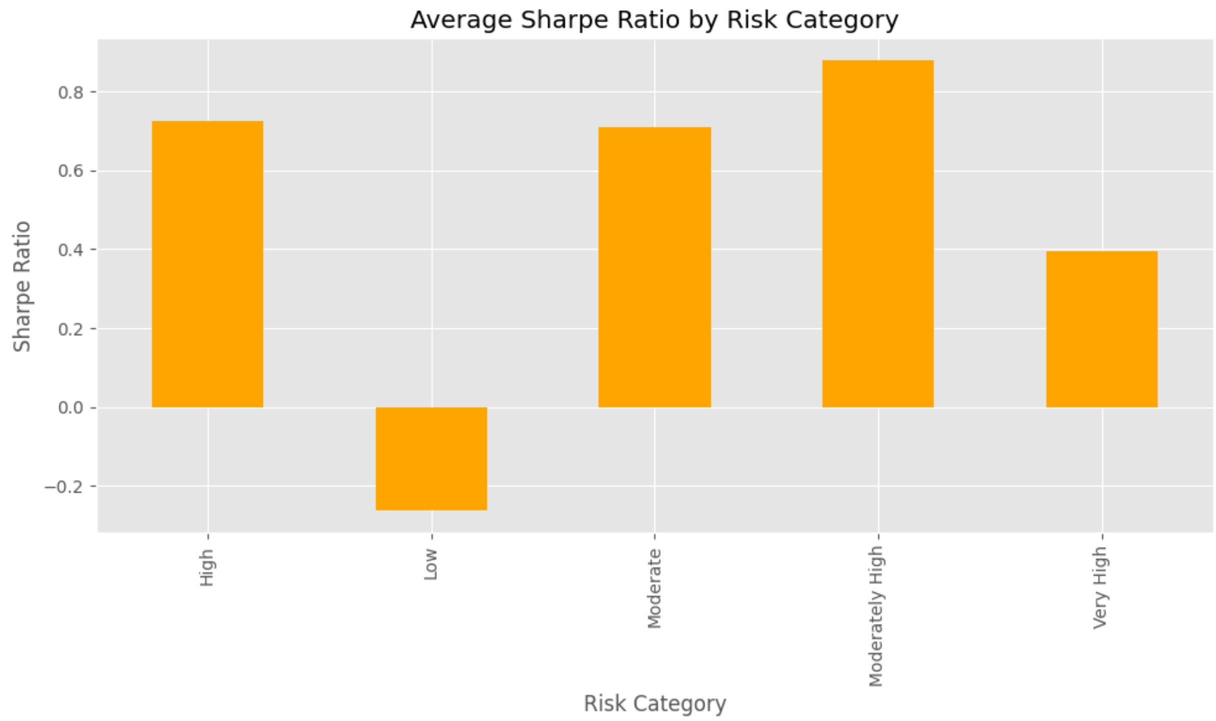


Figure 4.1b – Average Sharpe Ratio by Risk Category

Observation

- Mirae Asset Large Cap Fund recorded the single highest Sharpe Ratio (1.45) and Sortino Ratio (2.39) of all 40 schemes.
- A high Sharpe Ratio does not always coincide with the highest CAGR — Mirae Asset Large Cap Fund ranks outside the top-5 by CAGR but leads on risk-adjusted return.

4.2 Sortino Ratio Analysis

The Sortino Ratio (sortino_values.csv) refines the Sharpe Ratio by penalizing only downside volatility. Consistent with the Sharpe results, Mirae Asset Large Cap Fund led on Sortino Ratio (2.39), followed by other Moderate-risk Large Cap schemes, confirming that these funds manage downside risk more effectively than pure return figures alone would suggest.

Observation

- Higher Sortino Ratio indicates stronger downside protection specifically, rather than just lower overall volatility.
- Several Very High risk Small Cap schemes post respectable Sortino Ratios (~1.6–1.8) despite high headline volatility.

4.3 Alpha and Beta Analysis

Alpha and Beta (alpha_beta.csv) were computed against each scheme's stated benchmark. The ICICI Pru Midcap Fund posted the highest Alpha (0.2951) among the CAGR leaders, confirming genuine benchmark outperformance rather than simple market beta exposure. Beta values across the dataset cluster close to zero, indicating most schemes' daily return series in this sample show low direct sensitivity to their benchmark's daily moves over the observed window.

Observation

- Positive Alpha indicates outperformance versus the stated benchmark after adjusting for market exposure.

- Funds combining high Alpha with high Sharpe/Sortino (e.g. ICICI Pru Midcap, Mirae Asset Large Cap) represent the strongest all-round performers in this dataset.

4.4 Maximum Drawdown Analysis

Maximum Drawdown (max_drawdown.csv) measures the largest peak-to-trough decline for each scheme. The SBI Small Cap Fund – Direct Plan recorded the steepest drawdown of all 40 schemes at -52.57%, while low-risk Liquid and Gilt schemes recorded drawdowns of typically less than -5%.

Observation

- Small Cap and Sectoral/Thematic schemes show the deepest drawdowns, consistent with their Very High risk classification.
- Investors prioritizing capital protection should weight Maximum Drawdown alongside CAGR when short-listing schemes.

4.5 Risk vs Return Analysis

A consolidated risk-vs-return view (Alpha on the x-axis, Beta on the y-axis, sized by AUM) is presented on Dashboard Page 2 (Figure 7.2), where fund houses cluster into two visible groups: a lower-Alpha, tightly-clustered group of large diversified fund houses, and a small number of higher-Alpha outliers with materially larger bubble size.

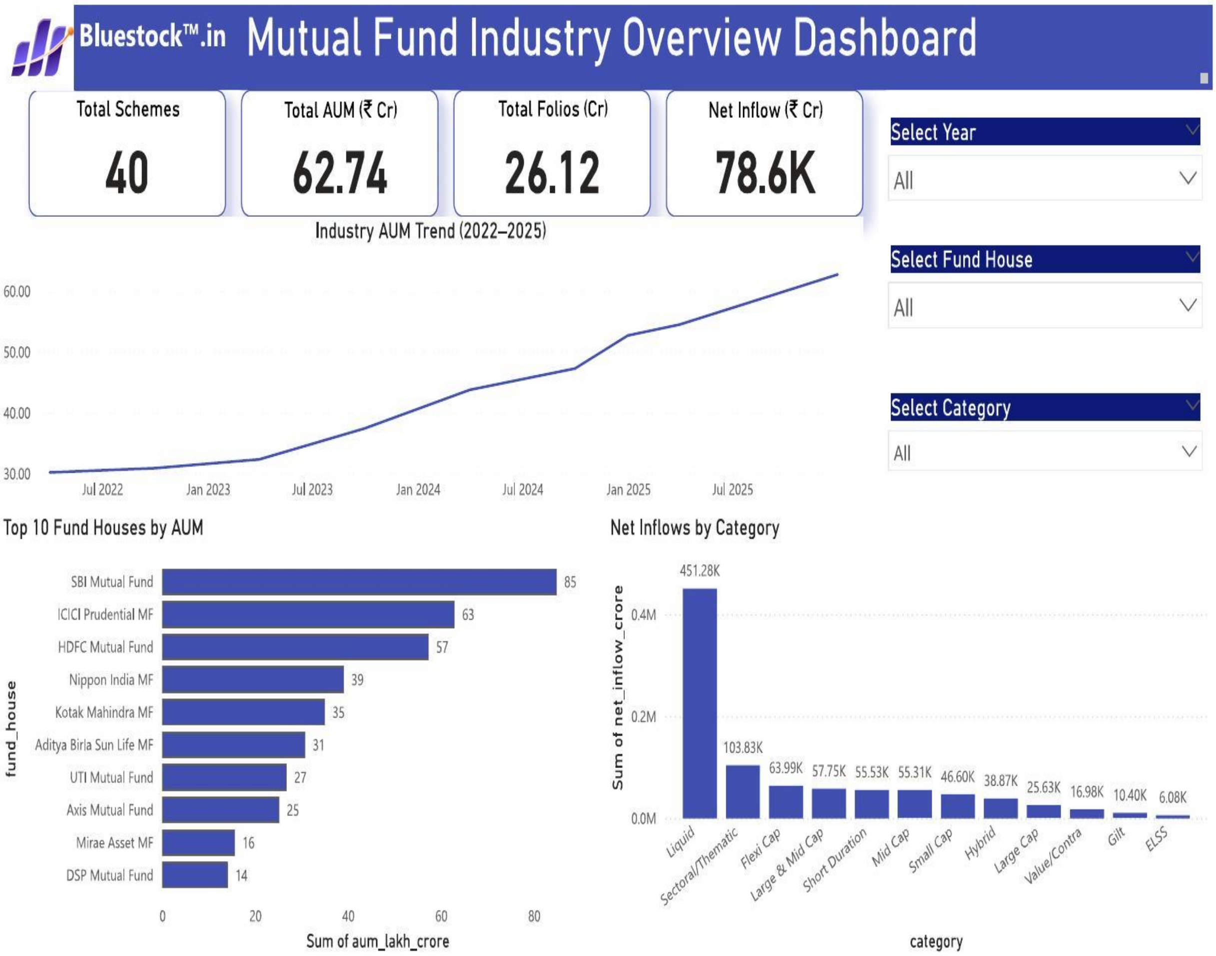
4.6 Key Findings

- ICICI Pru Midcap Fund led all 40 schemes on CAGR (32.83%) and ranked highly on Alpha (0.2951).
- Mirae Asset Large Cap Fund delivered the best risk-adjusted performance overall (Sharpe 1.45, Sortino 2.39).
- SBI Small Cap Fund – Direct Plan combined the weakest VaR/CVaR profile with the deepest Maximum Drawdown (-52.57%), making it the highest-risk scheme in the dataset by multiple measures.
- Debt and Liquid schemes remain the most defensive segment across every risk metric, at the cost of materially lower CAGR.

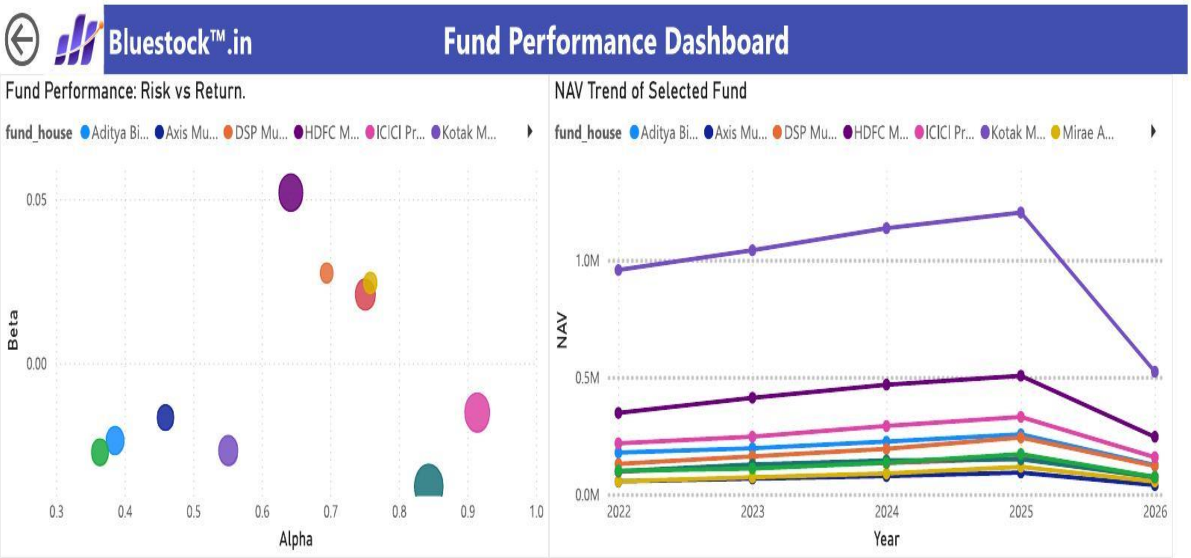
These findings carry directly into the advanced risk analytics performed in Chapter 6.

5. Dashboard Screenshots

A four-page interactive Power BI dashboard (bluestock_mf_dashboard.pbix) was built on the processed data to support business decision-making.



Page 1 – Industry Overview: 40 schemes, ₹62.74 Cr Total AUM, ₹78.6K Cr Net Inflow. SBI MF leads AUM; Liquid funds lead category inflow (451.28K).



Fund House

All

Category

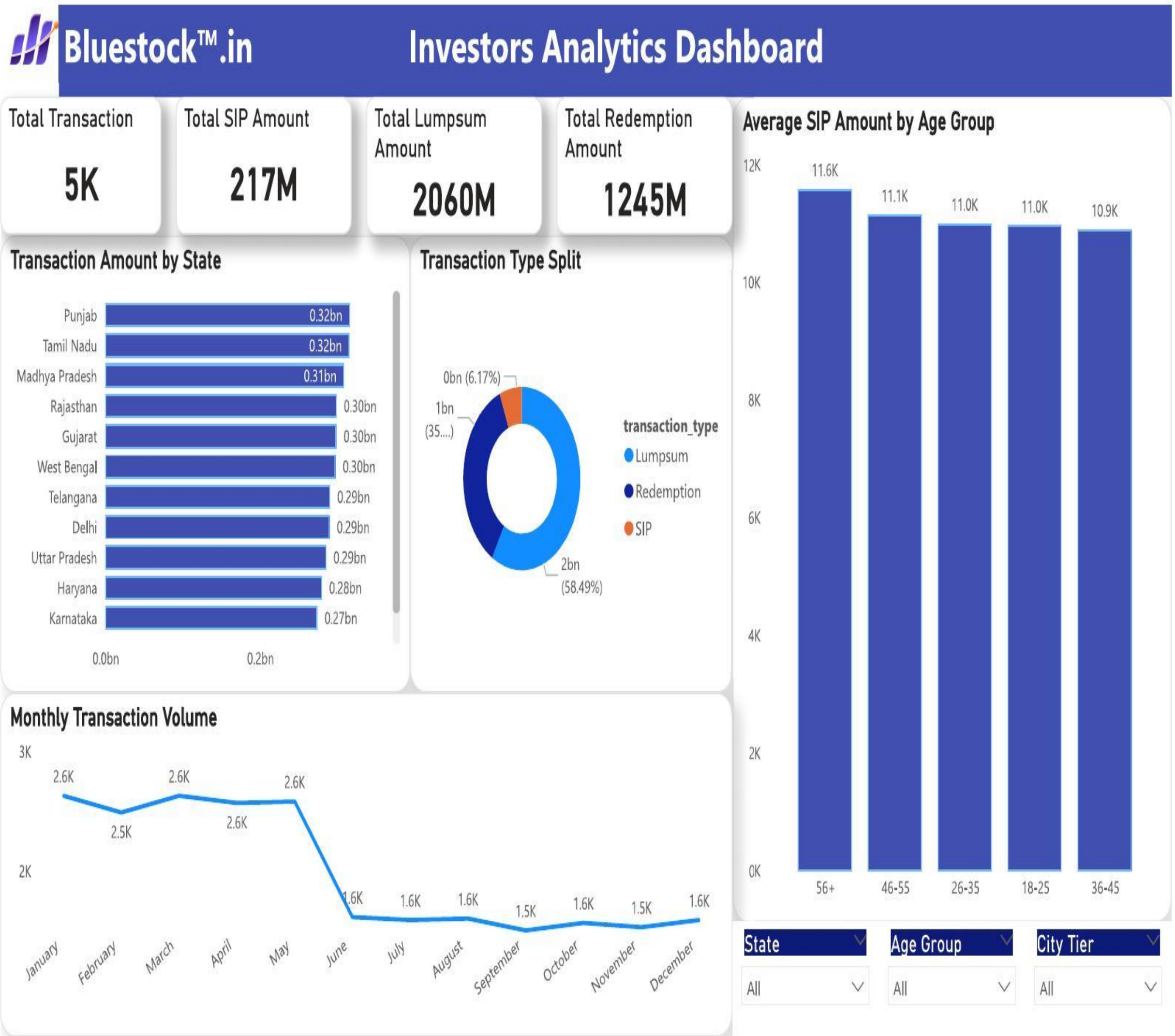
All

Plan

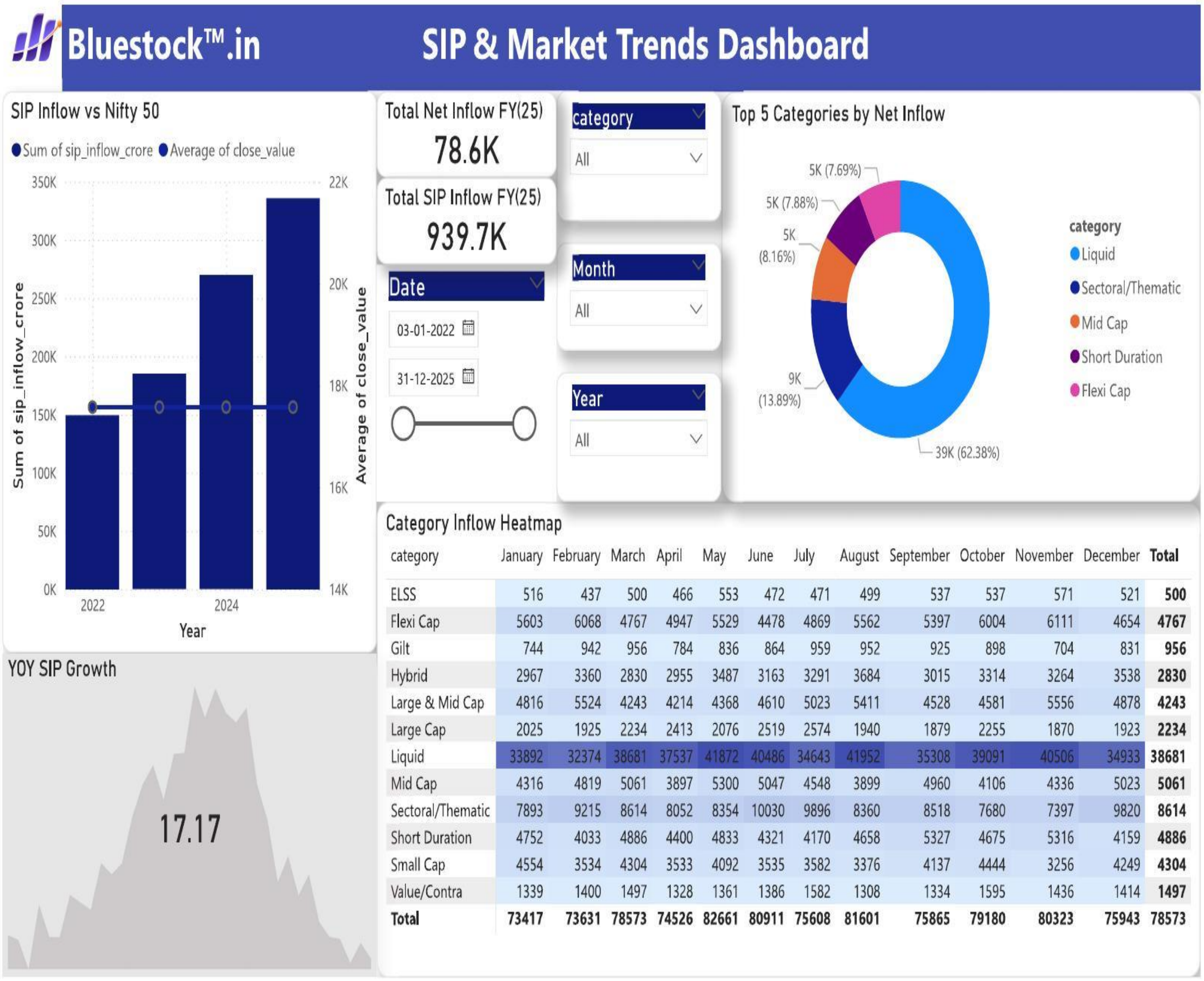
All

fund_house	Sum of Alpha	Sum of Beta	Sum of CAGR (%)	Sum of Sharpe Ratio	Sum of Sortino Ratio	Sum of Maximum Drawdown (%)
Mirae Asset MF	0.76	0.02	85.44	3.61	6.17	-42.94
Aditya Birla Sun Life MF	0.39	-0.02	37.99	0.37	0.40	-46.90
Kotak Mahindra MF	0.55	-0.03	58.06	1.79	3.15	-54.66
UTI Mutual Fund	0.36	-0.03	36.24	1.23	2.18	-60.40
DSP Mutual Fund	0.69	0.03	75.26	2.54	4.26	-63.26
ICICI Prudential MF	0.91	-0.02	99.24	4.16	7.33	-64.61
Nippon India MF	0.75	0.02	77.97	2.48	4.28	-79.35
HDFC Mutual Fund	0.64	0.05	67.54	1.40	2.38	-81.58
Axis Mutual Fund	0.46	-0.02	44.00	1.08	1.84	-108.79
SBI Mutual Fund	0.84	-0.04	87.68	2.82	4.96	-112.42
Total	**6.36**	**-0.02**	**669.42**	**21.48**	**36.95**	**-714.91**

Page 2 – Fund Performance: Alpha/Beta risk-return map, NAV trend, and Sharpe/Sortino/Drawdown by fund house.



Page 3 – Investors Analytics: 5K transactions, 58.5% Lumpsum share, SIP amount stable across age groups.



Page 4 – SIP & Market Trends: 17.17% YoY SIP growth; Liquid funds capture 62.38% of net inflow.

Chapter 6: Advanced Analytics & Risk Metrics

6.1 Introduction

Advanced Analytics extends the Chapter 5 metrics with statistical risk assessment, investor behavior analysis, and a rule-based recommendation system. Python, Pandas, NumPy and Matplotlib were used throughout, with results exported as CSV reports and chart images.

6.2 Historical Value at Risk (VaR)

Historical VaR at the 95% confidence level was calculated for each scheme as the 5th percentile of its daily NAV returns (var_cvar_report.csv), using the historical simulation approach.

Table 6.1 – Highest Risk Schemes by VaR (95%)

Scheme	Fund House	VaR 95%	CVaR 95%
SBI Small Cap Fund – Direct Plan	SBI Mutual Fund	-2.69%	-3.24%
Axis Small Cap Fund – Regular	Axis Mutual Fund	-2.62%	-3.17%
ABSL Small Cap Fund – Regular	Aditya Birla Sun Life MF	-2.60%	-3.25%
Nippon India Small Cap Fund – Regular	Nippon India MF	-2.54%	-3.23%
SBI Small Cap Fund – Regular Plan	SBI Mutual Fund	-2.45%	-3.06%

Table 6.2 – Lowest Risk Schemes by VaR (95%)

Scheme	Fund House	VaR 95%	CVaR 95%
ICICI Pru Liquid Fund – Regular	ICICI Prudential MF	-0.02%	-0.04%
ABSL Liquid Fund – Regular	Aditya Birla Sun Life MF	-0.03%	-0.04%
Kotak Liquid Fund – Regular	Kotak Mahindra MF	-0.03%	-0.04%
HDFC Short Term Debt Fund – Regular	HDFC Mutual Fund	-0.38%	-0.50%
Nippon India Gilt Securities Fund	Nippon India MF	-0.38%	-0.49%

Observation

- Small Cap equity schemes across every fund house consistently show the most negative VaR, meaning the largest expected 1-day loss under normal market conditions at 95% confidence.
- Liquid schemes show near-zero VaR, confirming their role as low-risk, cash-equivalent instruments.

6.3 Conditional Value at Risk (CVaR)

CVaR (Expected Shortfall) measures the average loss beyond the VaR threshold and is consistently more negative than VaR for every scheme in the dataset — for example, the SBI Small Cap Fund – Direct Plan shows a VaR of -2.69% but a CVaR of -3.24%, meaning that on the worst 5% of days, the average loss is materially larger than the VaR figure alone would suggest.

Observation

- CVaR provides a more conservative, tail-risk-aware view than VaR alone and should be preferred for Very High risk scheme evaluation.

6.4 Rolling 90-Day Sharpe Ratio

A 90-Day Rolling Sharpe Ratio was computed for a sample of five schemes (AMFI codes 100016, 100025, 100033, 101206, 101207) to observe how risk-adjusted performance evolves over time, rather than looking at a single static Sharpe figure.

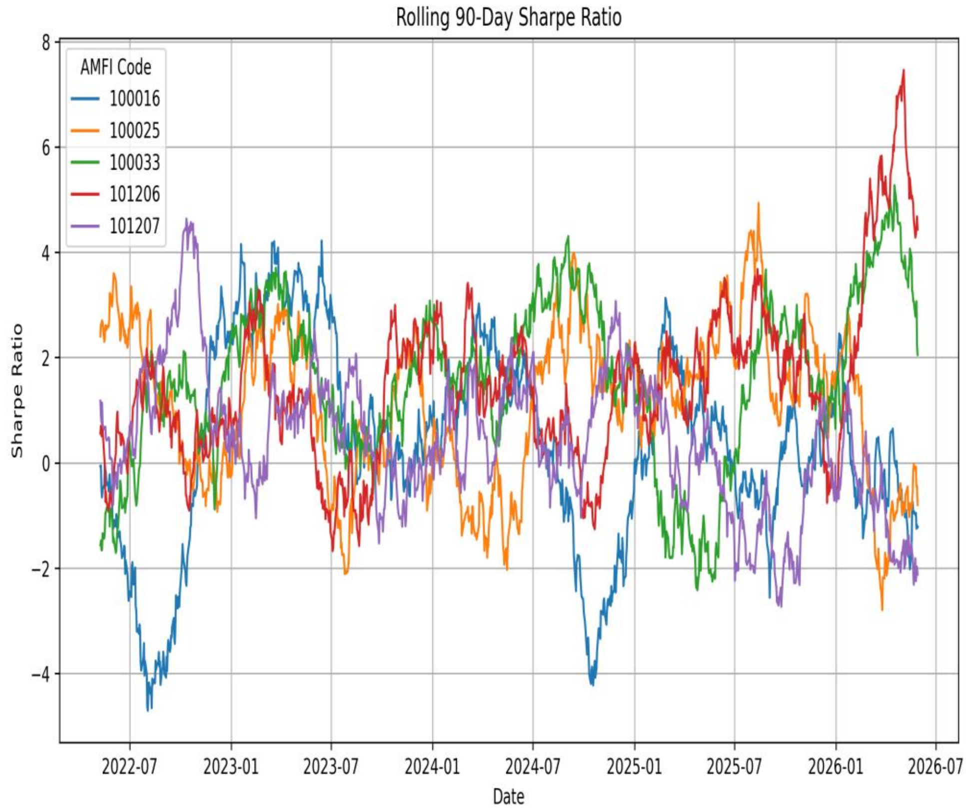


Figure 6.1 – Rolling 90-Day Sharpe Ratio (sample of 5 schemes, 2022–2026)

Observation

- Rolling Sharpe values swing between roughly -4 and +7 across the observed period for these five schemes, showing that short static Sharpe windows can be highly misleading.
- Scheme 101206 shows the strongest and most sustained improvement in rolling Sharpe through late 2025–2026.

6.5 Investor Cohort Analysis

Investors were grouped by the year of their first transaction and compared on average SIP amount, total invested, and investor count (cohort_analysis.csv).

Table 6.3 – Investor Cohort Summary

Cohort Year	Avg. SIP Amount (₹)	Total Invested (₹)	Total Investors
2024	1,07,422.54	3,49,11,25,187	4,803
2025	1,09,159.58	3,04,55,243	197

Observation

- The 2024 cohort is far larger (4,803 investors) and has contributed dramatically more total capital than the smaller, newer 2025 cohort (197 investors) — expected, given the shorter time window for the 2025 cohort to accumulate investments.
- Average SIP amount is marginally higher for the 2025 cohort (₹1,09,160 vs ₹1,07,423), suggesting slightly larger ticket sizes among newer investors in this sample.

6.6 SIP Continuity Analysis

SIP Continuity Analysis (sip_continuity.csv) calculated the average gap between consecutive SIP transactions for 1,362 tracked investors and classified each as Active or At Risk (average gap > 35 days).

Table 6.4 – SIP Continuity Status

Status	Investor Count	% of Total
At Risk (avg. gap > 35 days)	1,332	97.8%
Active	30	2.2%

The overall average gap between SIP contributions across all tracked investors is 64.9 days — nearly double the 35-day threshold used to flag risk.

Observation

- Almost all tracked investors (97.8%) show irregular SIP contribution patterns, representing a significant retention risk for fund houses relying on steady SIP inflows.
- This is one of the most actionable findings in the project — see Recommendation 2 in Chapter 8.

6.7 Mutual Fund Recommendation System

A rule-based recommendation engine (recommender.py) filters schemes by risk grade and ranks them by Sharpe Ratio, returning the Top 3 matches. Sample outputs from the processed dataset:

Table 6.5 – Sample Recommendations by Risk Grade

Risk Grade	Rank 1	Rank 2	Rank 3
Low	ICICI Pru Liquid Fund (Sharpe 0.50)	Kotak Liquid Fund (Sharpe -0.09)	SBI Magnum Gilt Fund (Sharpe -0.23)
Moderate	Mirae Asset Large Cap Fund (Sharpe 1.45)	SBI Bluechip Fund (Sharpe 1.21)	Nippon India Large Cap Fund (Sharpe 1.08)
Very High	SBI Small Cap Fund (Sharpe 0.95)	DSP Small Cap Fund (Sharpe 0.95)	Nippon India Small Cap Fund (Sharpe 0.45)

Observation

- Even within the Low risk grade, Sharpe Ratio can be negative — a reminder that risk grade alone does not guarantee a positive risk-adjusted return over every period.
- Sharpe Ratio was chosen as the ranking criterion because it directly measures return per unit of total risk, aligning recommendations with each investor's stated risk appetite.

6.8 Sector Concentration Analysis (HHI)

Sector concentration was evaluated using the Herfindahl-Hirschman Index (HHI = $\Sigma\, w_i^2$, where w_i is each sector's portfolio weight). Mean HHI across the 35 schemes analyzed was 0.140.

Table 6.6 – Most and Least Concentrated Portfolios

Scheme	Fund House	HHI	Interpretation
Axis Bluechip Fund – Regular	Axis Mutual Fund	0.206	Most concentrated
ABSL Small Cap Fund – Regular	Aditya Birla Sun Life MF	0.201	High concentration
SBI Small Cap Fund – Regular Plan	SBI Mutual Fund	0.107	Most diversified
SBI Bluechip Fund – Direct Plan	SBI Mutual Fund	0.108	Highly diversified

Observation

- Axis Bluechip Fund shows the highest sector concentration (HHI 0.206) among all analyzed schemes, indicating a comparatively narrow sector allocation.
- SBI Small Cap Fund – Regular Plan is the most sector-diversified scheme in the dataset (HHI 0.107), despite carrying a Very High overall risk grade — a useful reminder that concentration risk and volatility risk are distinct dimensions.

6.9 Advanced Business Insights

- Historical VaR/CVaR consistently identify Small Cap schemes as the highest downside-risk group across every fund house.
- Rolling Sharpe Ratio shows that risk-adjusted performance is highly time-varying — static, single-period Sharpe figures can be misleading for scheme selection.
- Investor Cohort Analysis shows the 2024 cohort dominates total invested capital, with limited data yet on 2025 cohort behavior.
- SIP Continuity Analysis reveals a major retention gap — 97.8% of tracked investors are classified At Risk.
- Sector HHI Analysis shows that risk grade and sector concentration are independent — a Very High risk scheme can still be well diversified by sector.

7. Recommendations

1. Promote High Risk-Adjusted Return Funds

Highlight Mirae Asset Large Cap Fund and ICICI Pru Midcap Fund to investors seeking long-term, risk-adjusted wealth creation, rather than promoting funds on headline return alone.

2. Address SIP Continuity Risk Urgently

With 97.8% of tracked investors classified At Risk (avg. 65-day contribution gap), deploy reminder notifications, auto-debit nudges, and targeted re-engagement campaigns for the At Risk segment.

3. Strengthen Diversification Monitoring

Review sector allocation for high-HHI schemes (Axis Bluechip Fund, ABSL Small Cap Fund) using SBI Small Cap Fund (HHI 0.107) as an internal diversification benchmark.

4. Use Risk-Adjusted Metrics Over Risk-Grade Labels Alone

Even Low-risk schemes can post negative Sharpe Ratios over a given period — advisors should reference Sharpe/Sortino/VaR figures, not risk grade alone, when recommending schemes.

5. Expand Dashboard & Model Capabilities

Add live NAV updates (building on live_nav_fetch.py), predictive NAV models, and a smarter recommendation layer that factors in VaR and HHI alongside Sharpe Ratio.

7. Limitations

While the platform delivers genuine, data-backed insights, the following constraints should be kept in mind when acting on its outputs:

- NAV gaps were forward-filled rather than sourced from an authoritative feed; this can understate volatility on days with genuinely missing market data.
- SIP Continuity and Cohort Analysis are based on a sample of 1,362 tracked investors, a subset of the full 32,000+ transaction records — findings should be validated against the full investor base before acting on them at scale.
- The 90-Day Rolling Sharpe Ratio was computed for a sample of 5 schemes only, not all 40 — it illustrates the method rather than providing scheme-by-scheme rolling risk for the full universe.
- Alpha and Beta are computed against a single stated benchmark per scheme; multi-factor or peer-relative benchmarking was out of scope.
- Historical VaR/CVaR assumes the historical return distribution is representative of future risk, which does not hold during unprecedented market events (e.g. a sudden liquidity crisis).
- The recommendation engine is rule-based (risk grade + Sharpe ranking only) rather than a trained model, and does not yet incorporate VaR, HHI, or investor-specific goals such as investment horizon.
- The Power BI dashboard reflects a static snapshot of the processed data rather than a live-refreshed connection to markets; figures should be refreshed periodically as new data arrives.
- SQLite was chosen for its simplicity and portability during development; it is not designed for concurrent multi-user or production-scale deployment, and a migration to PostgreSQL/cloud DB would be needed for production use.

- Sector HHI analysis covers 35 of the 40 schemes (portfolio holdings were unavailable for 5 schemes at the time of analysis).

8. Future Scope

- Integration of live mutual fund NAV data via the MFAPI endpoint already used in live_nav_fetch.py.
- Real-time dashboard refresh using Power BI Service.
- Machine learning models for NAV / return forecasting.
- A more sophisticated recommendation engine incorporating VaR and HHI alongside Sharpe Ratio.
- Portfolio optimization using Modern Portfolio Theory across the 40-scheme universe.
- Cloud deployment of the SQLite/ETL pipeline using Azure or AWS, with scheduled ETL via Apache Airflow.

9. Conclusion

The Mutual Fund Analytics Platform successfully demonstrated the complete lifecycle of a real-world data analytics project — from raw CSV ingestion and cleaning, through SQLite database design and SQL querying, to performance analytics, advanced risk metrics, and an interactive Power BI dashboard.

Across 40 schemes and 10 fund houses, the analysis identified ICICI Pru Midcap Fund and Mirae Asset Large Cap Fund as standout performers on CAGR/Alpha and Sharpe/Sortino respectively, flagged Small Cap schemes as the highest downside-risk group via VaR/CVaR and Maximum Drawdown, and surfaced a critical SIP continuity risk affecting nearly 98% of tracked investors.

Overall, this capstone project strengthened practical skills in Python, SQL, ETL pipeline development, data cleaning, exploratory data analysis, financial risk analytics, and Power BI dashboard development, while producing genuinely actionable, data-backed recommendations for both investors and fund managers.